



Thermal Decomposition Analysis of Soft Materials

Jean Kim, Advanced Polymeric Materials Branch Intern

Witold Fuchs, Ph.D Advanced Polymeric Materials
Branch, M&P Technical Lead



Bottom-line Up Front

- **Objective:** Investigate the mass loss of soft goods at project relevant conditions
- **Instruments:** Thermogravimetric Analysis (TGA) and vacuum oven
- **Samples:** Lacing Tape, Shrink Tubes

The mass loss increases with temperature in non-linear, exponentially increasing trend for both materials.



Instruments

Thermogravimetric Analysis (TGA)

- Continuous mass measurement during heating.
- Identifies thermal decomposition and oxidation behavior.
- Output: Mass loss vs. Temperature/Time

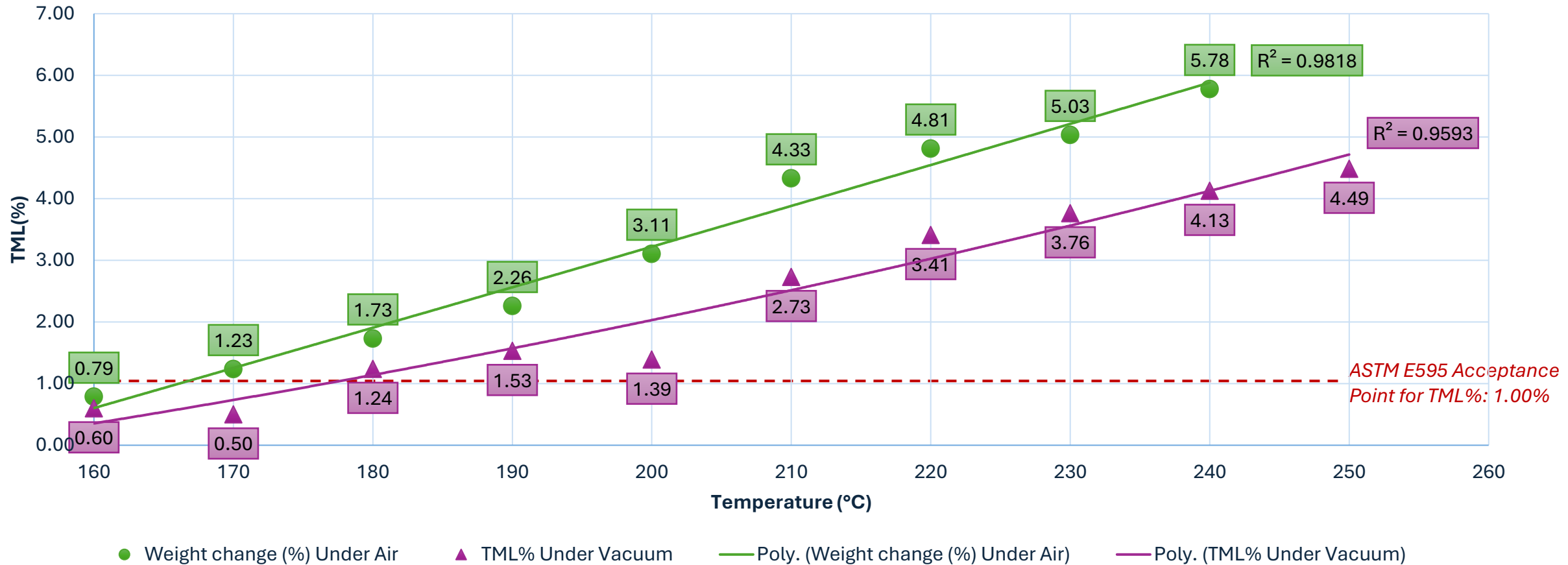
Vacuum Oven

- Endpoint measurement before and after exposure to heat.
- Vacuum level meet ASTM E595 standard
- Measures released volatile material under vacuum
- Output: TML

*TGA shows the progression of mass loss during heating.
TML reports the net volatile loss after a defined exposure to TVAC*



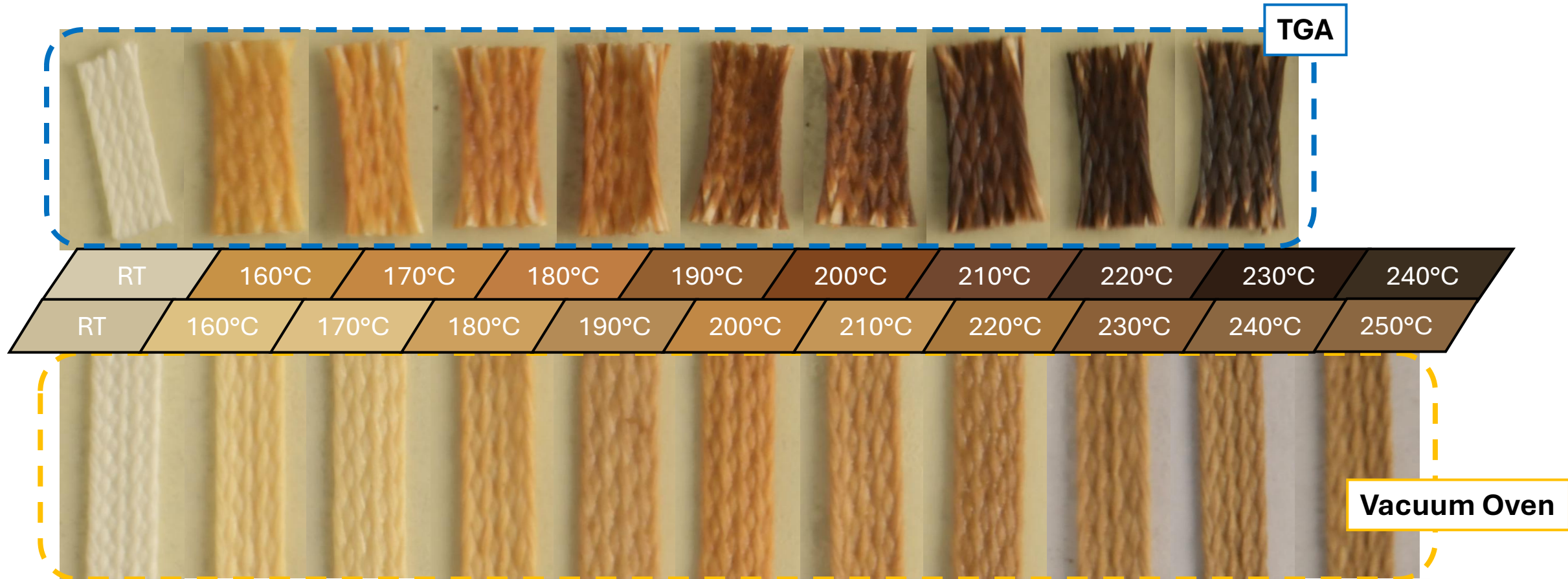
Total Mass Loss(%) vs. Temperature (°C) of Lacing Tape



The mass loss of lacing tape exceeds acceptance region of < 1.00% around 160-170°C



Colorimetric Analysis of Lacing Tape

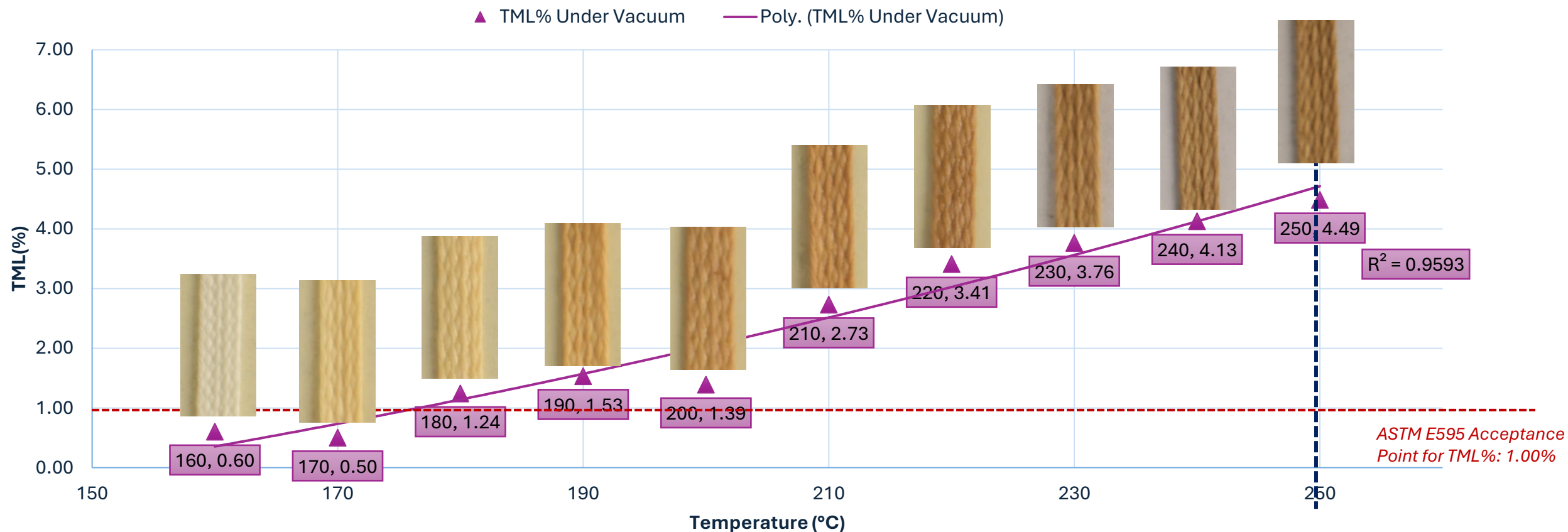


The most aged sample (240°C) is **64.8% darker** than the control (RT) sample.*



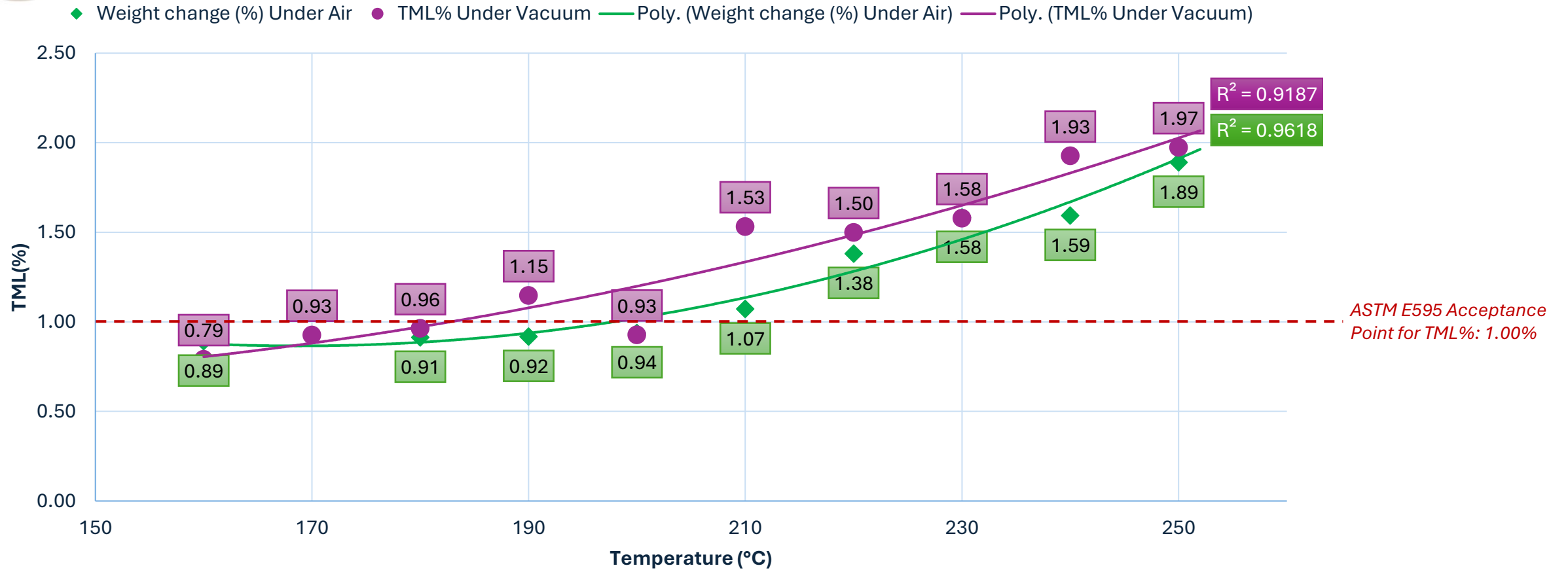
Surface Discoloration and TML%

Total Mass Loss(%) vs. Temperature (°C) **Lacing Tape** [Vacuum Oven]





Total Mass Loss(%) vs. Temperature (°C) of Shrink Tube

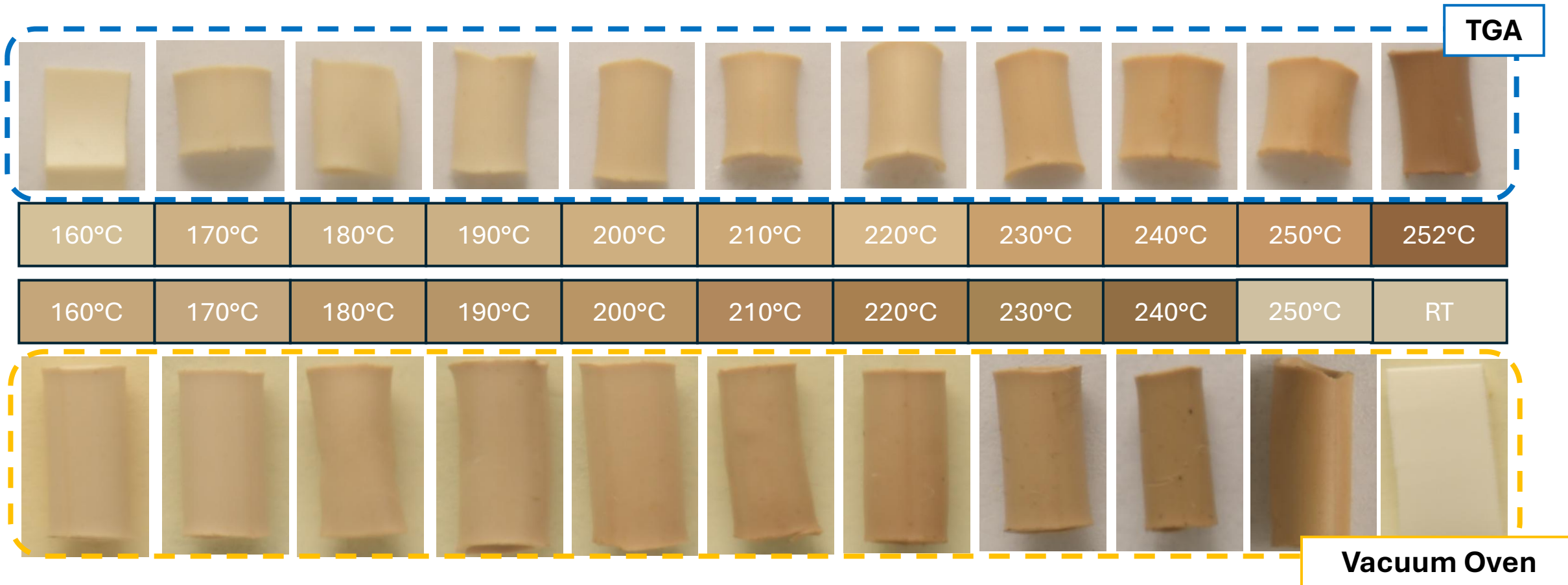


The mass loss of shrink tube exceeds acceptance region of < 1.00% around 200-210°C



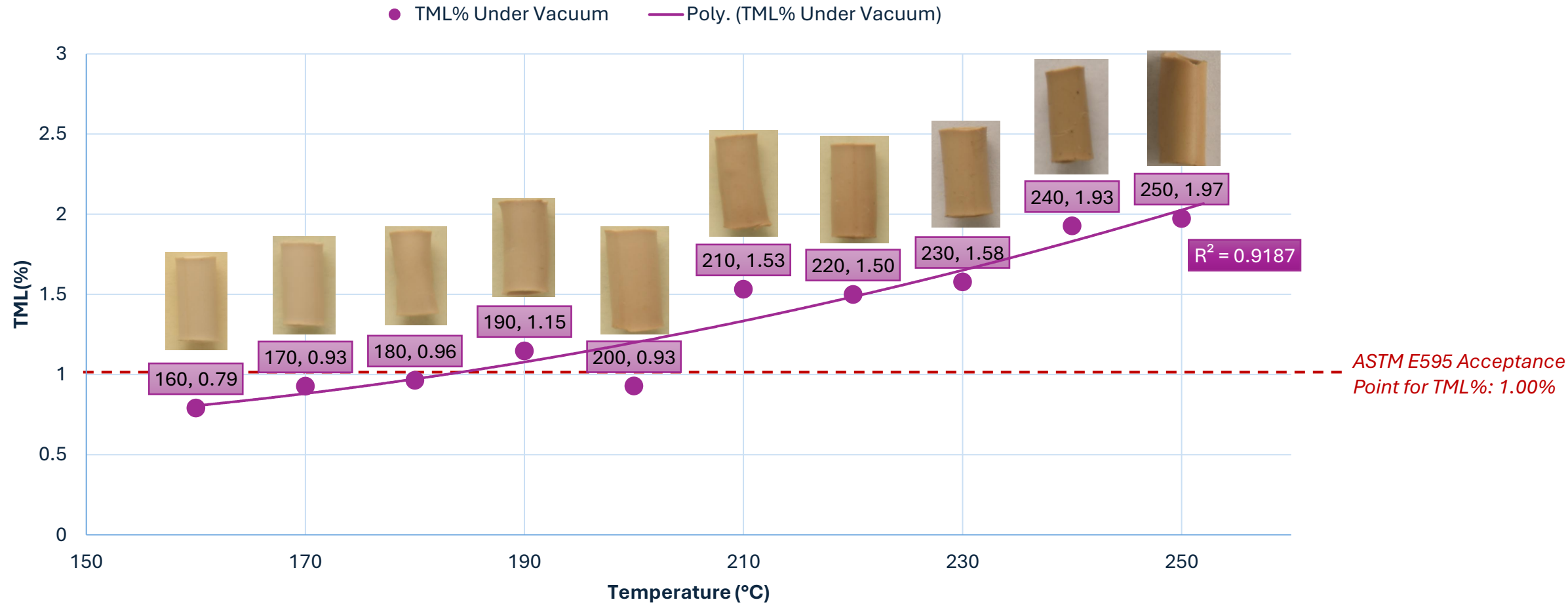
Shrink Tube

- Heat-shrinkable fluoropolymer tubing for high temperature applications.
- Operating temperature: -55°C to 225°C (-67°F to 437°F)





Total Mass Loss(%) vs. Temperature (°C) of Shrink Tube





Conclusion

- **Objective:** Investigate the mass loss of soft goods at project relevant test conditions using TGA and vacuum oven.

For both Lacing Tape AND Shrink tube, mass loss increases with temperature in non-linear, exponentially increasing trend.

Measured mass loss exceeds the 1.00% acceptance region at 160-170°C for lacing tape and 200-210°C for shrink tube in controlled setting.

Surface discoloration correlates with measured mass loss for both materials. It is plausible to assess the thermal degradation of soft goods from surface color provided that the image is taken under standardized lighting.